1 Discussion

What the formalism shows. Solomonoff axiom induction turns the view that theories answer questions posed in a given language into a precise object, and the view survives in comparative form. Its formal content is a change of target and of data type, logical coherence and partiality: consequences of the data are certain, "independent" is a native outcome, and a law is a sentence with a belief of its own (**????**). The inference principle is unchanged: version by version, SAI is operator induction over consistent labellers (**??**). What logic adds is a set of phenomena with no analogue for labellers: random theories that succeed where every r.e. theory fails, a single random true theory that is, up to a constant factor, the whole mixture on Π_1 data, and the fixed marks that Gödel's theorems leave on the posterior (**??**). On what a theory is, the idealised formalism is neutral: it cannot see the difference between a theory and a program. Where shape matters, it is relative to a reasoner and a vocabulary, and it matters only when the written theory is charged or the class of hypotheses is restricted (**????**). We still find it plausible that good theories are shaped like logical theories, and that textbook theories are declarative is an observation we accept; what is missing is a formal setting in which that shape provably pays. Finally, that Kepler's laws are only approximately true is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view: under SAI's pricing an approximately true law is a law with a calibrated error model (**??**).

What it cannot show. The axiom view models the declarative core of scientific theories: contents, inferential relations, partiality and approximate truth. It misses their procedural side (recognizers, methods, the search for derivations), the statistical confrontation of theory with data, of which its proof-requiring core is the noise-free special case, and their dynamic side (meaning change, discovery, the social process of science) (**??**). Its limit posterior can live on random theories and, with adversarially chosen true data, on false ones, so it supports no realist conclusion about the theories it favours (**????**). It evaluates theories and does not find them, and it is not computable.

1.1 Open problems

(1) Does M_A dominate every lower semicomputable provability function of a random consistent theory (**??**)?

(2) Is there a stream with bounded one-tape loss on which the coin-free loss diverges computably fast, and does bounded one-tape loss on a co-r.e. stream imply that a consistent r.e. theory proves it (**??**)?

(3) Over a true background containing PA, is the one-tape posterior probability of Σ_1-soundness bounded away from 1 uniformly over all data (**??**)?

(4) Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws, to real-valued laws with numerical solvers, probabilistic programs with conditioning and simulation codes (**??**)?

(5) Is there a formal setting for communities of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs (**??**)?

Further questions concern the random tape and the order of the data (????); the computability and the versions of consistent operator induction (????); Presburger data under charged deduction (??); soundness on true data, a coding theorem with constant slack and the loss on true arithmetic (??????); a Kreisel-type property for finite theories (??); the fit classes of richer observational languages, of infinite hypotheses about the named objects and of unary theoretical functions (??????); exact minima, learning and time bounds for usable theories (??????); and restricted classes of approximate laws and domains of applicability (????).

Directions. Four directions seem to us most worth pursuing. The first is step 4 of the ladder, the revision of the base abstractions, for which we have no formal handle. The second is the setting of the fifth problem above, which may also be where the declarative shape of written theories finds its explanation. The third is a costed-deduction mixture over unknown axiom systems, combining SAI, which is uncertain about the axioms, with logical induction (Garrabrant et al., 2016), which is uncertain about their consequences; such a reasoner is what induction in mathematics needs, since with free deduction checking instances confirms nothing (??). The fourth is a native treatment of statistical confrontation inside the proof-requiring core, beyond randomised theories and the generative variant (??). Smaller questions remain: whether explanatory asymmetry is interventional in general; a derivation of Kepler's laws within stated tolerances from Newton's theory in a prover for validated numerics, a concrete case of the computational gap of ??; and the Mars posterior with distances as data.

Acknowledgements

Several of the questions and examples studied here first appeared in Hänni's working notes (Hänni, 2026). Independent instances of Claude checked each part: they re-derived the proofs and re-implemented the computations, and the results were revised in response. The results of the earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion papers (Hänni and Claude (Anthropic), 2026b,c) that we use were prepared and checked in the same way. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant; readers should weigh the results accordingly. Hänni takes responsibility for the content of this paper.

Code and data

All computations reported here were done with seeded or deterministic Python scripts (Python 3.11 with NumPy, SciPy and mpmath; the planetary computations use pyerfa, the Python interface to the ERFA library). The scripts, their saved outputs and independent re-implementations of the main computations are in the folder `solomonoff-axiom-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each reported number and how to re-run it. Re-running the scripts reproduces the saved outputs, apart from lines that report run times; the longest computations, on the Mars pool, take more than half an hour each, and their figures were also reproduced by independent re-implementations.

A How the results were checked (internal record)

This appendix is an internal record. It is not part of the version of the paper meant for arXiv, where the account of checking is the title-page footnote and the acknowledgements.

A.1 The process

Research threads. The results come from seven research threads, each with a final record in the folder `research/tracks/<key>/` of the project: `fundamentals` (????), `sfi` (??), `logic` (??), `philosophy` (??????), `shape` (??), `terms` (??) and `laws` (??). Each thread was developed by one Claude instance from a brief that fixed the epistemic policy of the project (the paper asserts only what we believe, whoever proposed it). Every claim in a record carries a status: proved (a full proof in the record), proved given a cited result, computed (a seeded script and its saved output in the thread's `checks/` folder), known (with the level at which the reference was verified), proof sketch, conjecture, open, refuted (kept with its counterexample) or assessment (a philosophical judgement argued in the record).

Referees. Each thread record was refereed by two separate Claude instances (referee A and referee B; reports `referee-A.md`, `referee-B.md` next to the record). The referees re-derived the proofs and re-implemented the main computations with independent code (`referee_code/`, and `referee_A_code/` for `philosophy`). No report found a fatal issue. The major issues (2 in `fundamentals`, 2 in `sfi`, 3 in `logic`, 2 in `terms`, 11 in `shape`, 8 in `laws`, 14 in `philosophy`) were all handled in a revision round, whose results are the final records: statements were corrected or weakened, refuted claims were kept as refuted, and new checks were written. Examples: the consistency statement of **??** needs a true background (the version without it is refuted by $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$); the doubly exponential bound for reading (a1) on Presburger data was withdrawn and is open (**??**); the written construction behind **??** was rewritten and re-checked; a distance table of the Kepler computations was refuted and replaced.

Synthesis and final check. A synthesis assembled the seven final records into one map of results and a plan of the paper, quoting every statement with its record's status, and re-ran every developer script with a saved output on fresh copies: 43 of 47 reproduced (29 byte-identical, 14 identical apart from a line reporting the run time); the four longest Kepler computations were not re-run there, and their figures rest on the revision round's runs and on the referees' independent code. A separate final check compared every headline statement of the synthesis with the records and corrected 19 passages in which the synthesis had dropped a qualifier, a hypothesis or a status (for example, that only the companion paper's results that hold up to additive constants transfer, **??**; that declarative against procedural form costs only a constant factor once intermediate concepts are available and the reasoner can chain, **??**); it found no headline result false and no cited proof missing. It re-ran three key scripts (the one-tape finite stand-ins, the lottery, the Kepler residuals), with byte-identical outputs apart from a run-time line.

Writing. A planning editor fixed the notation, a ledger of every result with its label, source, status and binding wording, and the outline. Every cited reference was verified by web search, by a scout, a thread, a referee or the bibliography editor, and the bibliography editor recorded the level reached (bibliographic data, abstract, quotation, secondary source) in

paper/bib/bib-report.md; no page could be fetched, so no full text was read. Section writers then wrote the sections from the final records, each checking its statements against the record (hypotheses, quantifiers, constants, status) and logging every deviation (paper/logs/); several corrected the records' wording where it overstated what the proofs give, always by weakening (for example, the constant of ?? depends on the question family; the text lengths in the flagpole example of ??; the time-constructibility hypothesis of ??; the entries of ?? that are sizes of constructions). The front matter, this appendix and the full build were written last.

A.2 What was not checked

- Nothing was verified in a proof assistant.
- No human has independently checked all of the mathematics. The checking described above was done by Claude instances, which can share blind spots.
- References were verified only at the levels recorded in paper/bib/bib-report.md, from the text a search tool returned; the paper attributes content to a work only at that level. The definition of the modified Demski prior is known only from published descriptions and forum posts cited for attribution, so the comparison with it (??) is stated for the processes as defined in the paper.
- The novelty claims are at search level (??); for ?? the paper says only that we did not find it stated and that it is probably known to specialists.
- Results of the earlier report and of the companion papers that are cited by number (for example Lemma E.12 and Theorem 6.14 of the earlier report, used under Assumption (S)) were checked when those reports were written, not re-derived here. The assumptions (S), (S_{tape}) and (F1) of ??, and the two assumptions behind the conditional Presburger result (??), are not proved.
- Computations check finite analogues, constructions and numbers; they add no evidence for statements about universal machines beyond what the proofs give. Every arcminute figure of ?? is against the plan94 ephemeris, whose own error is quoted from its documentation.
- The philosophical conclusions are assessments, argued but not proved; ?? separates them from the theorems they rest on.

A.3 Status of every main-text result

The table lists every theorem-like environment, labelled paragraph, table and figure of the main text, with the status and source item that its \status and \src calls record (sources: <thread> <item> of the final record; "syn" is the synthesis, "fc" the final check). Items whose statement is in an appendix are not listed; their status is recorded in the same way in the appendix files.

item	label	status	source
setting.tex			
??	def:setting:data	definition	fundamentals Def 1.1
??	def:setting:fresh	definition	fundamentals Def 1.1'; terms Def 1.2
??	def:setting:machine	definition	fundamentals Def 1.2; Prop 2.1(g)
??	def:setting:MA	definition	fundamentals Def 1.4; NOTATION §4

item	label	status	source
??	def:setting:conventions	definition	fundamentals Def 1.5; logic §1.3
??	rem:setting:defects	proved	fundamentals Rem 1.6
??	def:setting:versions	definition	fundamentals Def 1.7
??	rem:setting:versions	(a), (c) proved; (b) proved, given Lemma E.12 and Theorem 6.14(a) of the earlier report, under Assumption (S)	fundamentals Prop 1.8(a)–(c)
??	def:setting:complexities	definition; the inequalities proved	fundamentals Def 1.9
??	def:setting:variants	definition	fundamentals Def 1.10; §7
??	def:setting:questions	definition	NOTATION §3.2; philosophy Def 2.0; shape Def 1.2, §1.3
basic.tex			
??	prop:basic:structure	proved; computed	fundamentals Prop 2.1
??	prop:basic:geometric	proved; computed; the names of the rules known	fundamentals Prop 2.2
??	prop:basic:dempster	proved; computed; the name of the rule known	fundamentals Prop 2.3
??	prop:basic:form	proved	fundamentals Prop 3.1
??	thm:basic:inc	(a), (b) proved; (c) proved, given Chaitin's theorem that Ω_V is Martin-Löf random and the Levin–Schnorr characterisation	fundamentals Thm 3.2
??	thm:basic:dominance	proved	fundamentals Thm 4.1
??	prop:basic:invariance	proved	fundamentals Prop 4.2
??	prop:basic:background	proved	fundamentals Prop 4.3
??	thm:basic:translation	proved	fundamentals Thm 4.4
??	cor:basic:vocabulary	proved	fundamentals Cor 4.5
??	thm:basic:loss	proved	fundamentals Thm 5.1
??	prop:basic:exceptions	proved	fundamentals Prop 5.2
??	prop:basic:mistakes	(a) proved; (b), (c) proved; computed	fundamentals Prop 5.3
??	thm:basic:merging	proved; computed	fundamentals Thm 5.4
??	cor:basic:threeloss	proved	fundamentals Cor 5.5
??	prop:basic:lower	proved	fundamentals Prop 5.6
??	prop:basic:chain	proved	fundamentals Prop 6.1
??	thm:basic:nocoding	proved; computed	fundamentals Thm 6.2
??	prop:basic:cr	proved; computed	fundamentals Prop 7.1
??	prop:basic:crorder	(a), (b) proved; computed	fundamentals Prop 7.2(a), (b)
??	prop:basic:prcr	proved	fundamentals Prop 10.1(f)
??	q:basic:domination	open	fundamentals Q6, Prop 11.4
soi.tex			
??	def:soi:objects	definition	sfi §1.1, Def 1.1
??	thm:soi:coinfree	proved; Craig's construction known	sfi Thm 2.1
??	thm:soi:onetape	proved; computed	sfi Thm 3.1
??	cor:soi:closed	proved; computed	sfi Cor 3.2
??	ass:soi:S	assumption	sfi §7 (the earlier report's (S))
??	ass:soi:Stape	assumption	sfi §3.3, Assumption (S_{tape})
??	thm:soi:penalty	(a) proved, under (S_{tape}); (b) proved, under (S)	sfi Thm 3.4 (a); Thm 7.1 (b)
??	thm:soi:transfer	proved; computed	sfi Thm 5.1
??	cor:soi:transfer	proved, by ?? and the cited results of the companion paper	sfi Cor 5.2, §5.5
??	rem:soi:encoding	proved; computed	sfi Rem 5.4
??	prop:soi:closure	proved; (a), (b), (d) computed	sfi Prop 6.1
??	prop:soi:independent	proved, given the productivity lemma (classical) and, one-tape, the Lebesgue density theorem; computed	sfi Props 6.3, 6.5
??	prop:soi:noRE	proved	sfi Prop 6.4
??	cor:soi:onesided	proved; for the penalised versions under (S)	sfi Cor 7.3′; Thm 7.8(a)
??	thm:soi:diagonal	proved, given Kleene's recursion theorem and the MRDP theorem	sfi Thm 7.4

item	label	status	source
??	prop:soi:freededuction	proved	sfi Prop 7.6 (a), (b)
??	rem:soi:timereading	assessment	sfi Rem 7.9
??	ass:soi:F1	assumption	sfi §8 (the earlier report's (F1))
??	thm:soi:sandwich	proved, under (F1); the correspondence of proof systems with NP known	sfi Thm 8.1
??	rem:soi:whatcharged	assessment	sfi Rem 8.5
??	prop:soi:presburger	proved, given Presburger's theorem, the Fischer–Rabin bound and, for (c), the Lebesgue density theorem	sfi Props 9.1, 9.2
??	prop:soi:presburgerCharged	proved, given the Fischer–Rabin bound, Presburger's theorem, Lemma E.28, Proposition E.23(a) and the proof of Theorem 6.9 of the earlier report	sfi Props 9.4, 9.4′
??	rem:soi:presburgerSummary	assessment (summary of proved results; the (a1) correspondence is a reading for sentence data)	sfi Rem 9.5
??	tab:soi	see the items it collects	–
logic.tex			
??	thm:logic:lottery	proved; computed	logic Thm 2.2
??	prop:logic:arity	proved; computed	logic Prop 2.3
??	cor:logic:Q1	proved	logic Cor 2.5
??	q:logic:computablefast	open	logic E-Q2, §2.6
??	thm:logic:arith	proved, given ??; (b) computed	logic Thm 2.9
??	cor:logic:false	proved; computed	logic Cor 2.10
??	thm:logic:threshold	proved; computed	logic Thm 3.2
??	cor:logic:coding	proved	logic Cor 3.3
??	cor:logic:chaitin	proved	logic Cor 3.6
??	prop:logic:normalised	proved, given (Uniq); computed	logic Prop 3.11
??	prop:logic:shape	proved; computed	logic Prop 4.1
??	thm:logic:soundness	proved, given (G2)	logic Thm 4.2
??	thm:logic:soundnessPi1	proved, given (G2); computed	logic Thm 4.4
??	q:logic:soundness	open	logic D-Q1, §4.4
??	prop:logic:oldevidence	proved	logic Prop 4.6
??	prop:logic:omega	proved	logic Prop 5.1
??	thm:logic:goedelomega	proved, given (G2)	logic Thm 5.3
??	rem:logic:closures	(a) proved; (b) proved, given Dedekind's categoricity theorem; (c) known	logic §5.4
??	thm:logic:thn	proved, given that Ω_V is Martin-Löf random; the exponent computed	logic Thm 6.1
??	cor:logic:undecided	proved	logic Cor 6.2
??	q:logic:thn	open	logic Q3, §6.2
phil.tex			
??	tab:phil:map	see the items it collects	–
??	q:phil:declarative	open	syn §6 item 26; philosophy §1.2, §5 (L11)
??	def:phil:questions	definition	philosophy Def 2.0
??	prop:phil:answers	proved; computed	philosophy Prop 2.1
??	prop:phil:extractor	proved, given thm:basic:dominance and the standard sampling construction	philosophy Prop 2.4
??	rem:phil:practice	assessment	philosophy §1.3 (o4), §2.6; syn §2.2 row D
??	rem:phil:purpose	assessment	philosophy §2.2 remark; syn §7 item 2
??	rem:phil:craig	assessment, resting on thm:soi:coinfree, thm:soi:onetape, cor:basic:vocabulary, thm:terms:horn, thm:terms:KCV, thm:shape:invisible, prop:phil:flagpolesystem, prop:soi:freededuction (proved)	syn §2.8 row 1; philosophy §1.4 item 1
??	prop:phil:flagpole	proved; computed	philosophy Prop 3.1
??	tab:phil:differences	see the items it collects	–

item	label	status	source
shape.tex			
??	def:shape:model	definition	shape Defs 1.1–1.4, 1.6
??	thm:shape:invisible	proved; computed	shape Thm 1.11
??	thm:shape:visible	proved	shape Thm 1.12
??	prop:shape:onlyforced	proved; computed	shape Prop 2.5a
??	thm:shape:minimum	proved; computed	shape Thm 2.6
??	cor:shape:simplex	proved; computed	shape Cor 2.8
??	prop:shape:oneround	proved; computed	shape Prop 3.1a
??	prop:shape:depth	(a), (b) proved; computed; (c), (d) proved, given Hastad 1986	shape Prop 3.8
??	thm:shape:systems	proved, given Goos–Kamath–Robere–Sokolov 2019; computed	shape Thm 3.6
??	cor:shape:regimes	proved, given Hastad 1986 and Goos–Kamath–Robere–Sokolov 2019 as indicated; the reading: assessment	shape Cor 3.9
??	tab:shape:regimes	see the items it collects	–
??	q:shape:beyond	open	synthesis 6 item 17; philosophy 1.4 item 2; shape 10
??	rem:shape:reading	assessment, resting on cor:shape:regimes, thm:shape:invisible, thm:shape:systems	shape 10; philosophy 5.1
terms.tex			
??	def:terms:obs	definition	terms Def 1.1
??	def:terms:literal	definition	terms Def 1.2
??	def:terms:named	definition	terms Def 1.3
??	thm:terms:dilemma	proved, with the conditions of the parts listed in the proof	terms Thm 8.1
??	thm:terms:horn	(a), (b), (d) proved; computed (a finite analogue of (d)); (c) proved, under (S) and (S_{tape}), given Lemma E.12 and Theorem 6.14 of the earlier report, thm:soi:penalty and prop:soi:classes	terms Thm 2.1
??	rem:terms:hempel	proved (the comparison); the reading is an assessment	terms Rem 2.2
??	thm:terms:padded	proved; (b) is Lemma E.22 of the earlier report in this form	terms Thm 3.3
??	thm:terms:regular	proved, given the Ehrenfeucht–Fraisse theorem	terms Thm 5.1
??	rem:phil:hempel	assessment, resting on thm:terms:horn and thm:terms:dilemma	philosophy §1.6; synthesis §2.4
laws.tex			
??	def:laws:setting	definition	laws Defs 1.1–1.3, 1.5, 1.7; §2 setting; Setting S3
??	prop:laws:rounding	proved; computed	laws Prop 2.5
??	prop:laws:errormodel	proved; computed	laws Prop 3.6a, Def 3.6a
??	rem:laws:typical	computed	laws Rem 4.4
??	prop:laws:kink	proved, given Lemma 3.8 and Corollary 5.4 of the companion paper on steeper priors; computed	laws Prop 5.6
??	def:laws:EP	definition	laws Def 5.2
??	rem:laws:domainlearning	assessment	laws §8
kepler.tex			
??	def:kepler:language	definition	philosophy §4.1
??	prop:kepler:helio	proved	philosophy Prop 4.2
??	prop:kepler:geo	proved; computed	philosophy Prop 4.3
??	prop:kepler:K3worth	proved; computed	philosophy Prop 4.4
??	ex:kepler:K3cost	computed (conditional on the code; the extrapolation and the compact-code threshold are estimates)	philosophy §4.4 Part B; laws Prop 5.5(b)
??	tab:kepler:minimax	see the items it collects	–
??	tab:kepler:map	see the items it collects	–
??	fig:kepler:map	see the items it collects	–

item	label	status	source
??	ex:kepler:graded	computed (the honest-band crossovers are estimates from the pricing of section 9)	philosophy §4.4 Part A'
limits.tex			
??	tab:limits:concepts	see the items it collects	–

References

Scott Garrabrant, Tsvi Benson-Tilsen, Andrew Critch, Nate Soares, and Jessica Taylor. Logical induction. arXiv:1609.03543, 2016.

Kaarel Hänni. Working notes, 2023–2026. Public repository, https://github.com/kaarelh/notes, 2026. Accessed October 2026. The note on Solomonoff axiom induction is in the folder ai/learning/solomonoff induction.

Kaarel Hänni and Claude (Anthropic). Bayesian axiom induction from instances: Template priors, derivation likelihoods and time penalties. Report, 2026a. Folder axiom-induction of https://github.com/kaarelh/AI-works.

Kaarel Hänni and Claude (Anthropic). Solomonoff operator induction: Definitions, versions and basic theorems. Preprint, 2026b. Folder solomonoff-operator-induction of https://github.com/kaarelh/AI-works.

Kaarel Hänni and Claude (Anthropic). Steeper simplicity priors in Solomonoff operator induction: The model–randomness split and the teacher's mistakes. Preprint, 2026c. Folder solomonoff-operator-induction of https://github.com/kaarelh/AI-works.